Probabilistic Methods

Pictures at an Exhibition

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The probabilistic method is a powerful technique that allows us to prove the existence of combinatorial objects with certain properties. The idea behind the method is very simple: if you want to show that a set of combinatorial objects A is non-empty, construct a probability space that contains A as an event and show that the probability of A is positive.

There is no better way to understand the probabilistic method than by seeing it in action. Let us start with an example that was studied by Paul Erdős, the man who is often credited with the invention of the probabilistic method.

Exercise 0.1 (Property B)**.**

1. A hypergraph $H = (V, E)$ is called r-uniform if each hyperedge $e \in E$ has size $|e| = r$. Show that every r-uniform hypergraph with $|E| < 2^{r-1}$ is 2-vertext-colorable.
2. (Erdös 1947) Let $G = (V, E)$ be a graph. Show that for $k \geq 3$, if $|V| < 2^{\frac{k}{2}}$, then there exists a 2-edge-coloring of G such that there is no monochromatic k-clique.
3. (ERDÖS 1963) Consider a family of sets $F_1, F_2, \dots, F_m$ such that $|F_i| = r$ for all i, and let $\mathcal{F} = \bigcup_{i=1}^{m} F_i$. A set $S \subseteq \mathcal{F}$ is said to have *property B* if for every i, $S \cap F_i \neq \emptyset$ and $F_i \not\subset S$. What condition on m ensures that there exists a set S with property B?
4. Can Part 2 be reduced to Part 3?

Solution

1. Color the vertices of H independently and uniformly at random by two colors. Call this coloring $\mathcal{C}$. We call a 2-vertex-coloring of H *proper* if no hyperedge is monochromatic. We have

$$\mathbb{P}[\mathcal{C} \text{ is a proper 2-coloring}] = 1 - \mathbb{P}[\text{a monochromatic hyperedge exists}],$$

and

$$\mathbb{P}[\text{a monochromatic hyperedge exists}] \leq \sum_{e \in E} \mathbb{P}[\mathbb{1}_{e \text{ is monochromatic}}].$$

Since H is r-uniform, the probability that a hyperedge e is monochromatic is 2^{-r+1}. Thus, we have

$$\mathbb{P}[\mathcal{C} \text{ is a proper 2-coloring}] \geq 1 - |E| 2^{-r+1}.$$

Since $|E| < 2^{r-1}$, this probability is positive, which implies that there exists a proper 2-coloring of H.

2. Color the edges of G independently and uniformly at random by two colors. Call this coloring $\mathcal{C}$, and let us call a coloring *good* if the colored graph has no monochromatic k-clique. We have

$$\mathbb{P}[\mathcal{C} \text{ is good}] = 1 - \mathbb{P}[\text{a monochromatic } k\text{-clique exists}].$$

We use the union bound to bound the probability of the existence of a monochromatic k-clique. Let us consider an arbitrary order for the k-cliques of G and denote them by $K_k^1, K_k^2, \dots, K_k^m$.

$$
\begin{aligned}
\mathbb{P}[\text{a monochromatic } k\text{-clique exists}] &\leq \sum_{i=1}^{m} \mathbb{P}[\mathbb{1}_{K_k^i \text{ is monochromatic}}] && (1) \\
&\leq m 2^{-k+1} && (2) \\
&\leq \binom{|V|}{k} 2^{-k+1} && (3) \\
&\leq \frac{|V|^k}{k!} 2^{-\binom{k}{2}+1} && (4) \\
&< 2^{k^2/2-\binom{k}{2}+1} < 1. && (5)
\end{aligned}
$$

Thus, the probability of $\mathcal{C}$ being good is positive, yielding the existence of a good coloring of the edges of G.

3. Reduce the problem to the first part.

The First Moment Method

Again, the idea is super easy: if you average over a set of numbers, there must be a number in the set that is at least as large as the average, and there has to be a number that is at most as large as the average. In mathematical terms, if X is a non-negative random variable with finite expectation μ, then

$$\mathbb{P}[X \geq \mu] > 0 \quad \text{and} \quad \mathbb{P}[X \leq \mu] > 0.$$

Exercise 0.1 (Finding a Large Cut in a Graph)**.**

1. Let $G = (V, E)$ be a graph with $|E| = m$. Show that there exists a cut $(S, V \quad S)$ with size at least $m/2$.
2. We now want to convert our existence proof into a randomized algorithm. Describe a Monte Carlo algorithm that finds a cut of size at least $m/2$.

3. To analyze the error probability of the algorithm, give a lower bound on the probability that the random cut you have defined in Part 1 has size at least $m/2$.

4. Revise the algorithm to make it a polynomial-time randomized algorithm with an exponentially small error probability.

Solution

1. Indepedently and uniformly at random, assign each vertex to either S or $V\ \ S$. Let X be the number of edges crossing the cut $(S, V\ \ S)$. We have

$$X = \sum_{e \in E} \mathbb{1}_{e \text{ crosses the cut}}.$$

Thus, the expected size of the cut is

$$\mathbb{E}[X] = \sum_{e \in E} \mathbb{P}[e \text{ crosses the cut}] = m/2.$$

By the first moment method, $\mathbb{P}[X \geq m/2] > 0$, which implies the existence of a cut of size at least $m/2$.

2. Consider the following Monte Carlo algorithm:

3. For $i = 1, 2, \dots, k$:

    1. Assign each vertex to S or $V\ \ S$ independently and uniformly at random.
    2. If the cut $(S, V\ \ S)$ has size at least $m/2$, return it.

4. Return “failure”.

Bibliography

Erdös, P. 1947. “Some remarks on the theory of graphs.” *Bulletin of the American Mathematical Society* 53 (4): 292–94.

ERDÖS, P. 1963. "On a Combinatorial Problem." *Nordisk Matematisk Tidskrift* 11 (1): 5–10. http://www.jstor.org/stable/24524364.